Tutorial 4, Problem 15

MH1200 Linear Algebra I

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Ex. 15. (i) Consider the matrix:

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

Compute $P^3 - I_3$. Show your working.

(ii) Let a and b be real numbers. Calculate the following quantity:

$$\mathrm{tr}\left(\begin{pmatrix} a & 0 & b \\ b & a & 0 \\ 0 & b & a \end{pmatrix}^9 \right)$$

(iii) Let a and b be real numbers. Calculate the following quantity:

$$\mathrm{tr}\left(\begin{pmatrix} a & 0 & b \\ b & a & 0 \\ 0 & b & a \end{pmatrix}^{100} \right)$$

Sol 15. In the interest of time, I will only solve (iii). We show a generalization of this property for arbitrary positive integer exponents n.

Define a permutation matrix as a matrix where each row is composed of all 0s except for exactly one entry, which is 1. We note that $P^3 = I_3$. An easy way to see this is to consider the isomorphism between the group consisting of permutation matrices of order n under the operation of matrix multiplication and the symmetric group over n elements, S_n. A more elementary argument is presented here:

Since $P^3 = I_3$, I claim that the two matrices commute with one another.

Notice that,

$$\begin{aligned} I_3 P &= P^3 P \\ &= P^4 \\ &= P P^3 \\ &= P I_3 \end{aligned}$$

Since the matrices commute with each other, they satisfy the binomial theorem.

Thus, we have:

$$\mathrm{tr}\left(\begin{pmatrix} a & 0 & b \\ b & a & 0 \\ 0 & b & a \end{pmatrix}^n \right) = \mathrm{tr}\left((aI_3 + bP)^n \right)$$

$$\begin{aligned}&=\mathrm{tr}\big((aP^3+bP)^n\big)\\&=\mathrm{tr}\Big(\textstyle\sum_{i=0}^{n}\binom{n}{i}a^ib^{n-i}P^{3i}P^{n-i}\Big)\\&=\mathrm{tr}\Big(\textstyle\sum_{i=0}^{n}\binom{n}{i}a^ib^{n-i}P^{n+2i}\Big)\end{aligned}$$

Note that $\mathrm{tr}\big(P^{0 \bmod 3}\big) = 3$, and $\mathrm{tr}\big(P^{1 \bmod 3}\big) = \mathrm{tr}\big(P^{2 \bmod 3}\big) = 0$. Therefore,

$$\begin{aligned}\mathrm{tr}\left(\begin{pmatrix}a&0&b\\b&a&0\\0&b&a\end{pmatrix}^n\right)&=\mathrm{tr}\Big(\textstyle\sum_{i=0}^{n}\binom{n}{i}a^ib^{n-i}P^{n+2i}\Big)\\&=\mathrm{tr}\Big(\textstyle\sum_{i=0,i\equiv n \bmod 3}^{n}\binom{n}{i}a^ib^{n-i}P^{n+2i}\Big)\\&=3\textstyle\sum_{j=0,j\equiv n \bmod 3}^{n}\binom{n}{j}a^jb^{n-j}\end{aligned}$$

To simplify this sum, we will use generating functions and complex numbers. The appearance of the $n \bmod 3$ restriction motivates the introduction of complex numbers to handle the cycling behavior of modular arithmetic.

Let $P(x) = \sum_{j=0}^{n}\binom{n}{j}a^jb^{n-j}x^j = (ax+b)^n$ be the ordinary power series generating function describing our sequence. Further, let $\zeta_1, \zeta_2, \zeta_3$ be the three roots of $x^3 - 1 = 0$ (i.e., the third roots of unity).

Notice that $\zeta_1^n + \zeta_2^n + \zeta_3^n = \begin{cases}0 \text{ if } n\not\equiv 0 \bmod 3\\3 \text{ if } n\equiv 0 \bmod 3\end{cases}$. Therefore, we have:

$$\begin{aligned}\mathrm{tr}\left(\begin{pmatrix}a&0&b\\b&a&0\\0&b&a\end{pmatrix}^n\right)&=3\textstyle\sum_{j=0,j\equiv n \bmod 3}^{n}\binom{n}{j}a^jb^{n-j}\\&=\zeta_1^{2n}P(\zeta_1)+\zeta_2^{2n}P(\zeta_2)+\zeta_3^{2n}P(\zeta_3)\\&=(a\zeta_1+b)^n+\zeta_2^{2n}(a\zeta_2+b)^n+\zeta_3^{2n}(a\zeta_3+b)^n\\&=(a+b)^n+\left(\tfrac{-1+i\sqrt{3}}{2}\right)^{2n}\left(a\tfrac{-1+i\sqrt{3}}{2}+b\right)^n+\left(\tfrac{-1-i\sqrt{3}}{2}\right)^{2n}\left(a\tfrac{-1-i\sqrt{3}}{2}+b\right)^n\\&=(a+b)^n+\left(a+\left(\tfrac{-1-i\sqrt{3}}{2}\right)b\right)^n+\left(a+\left(\tfrac{-1+i\sqrt{3}}{2}\right)b\right)^n\end{aligned}$$

Thus, we have:

$$\mathrm{tr}\left(\begin{pmatrix}a&0&b\\b&a&0\\0&b&a\end{pmatrix}^n\right) = (a+b)^n + \left(a+\left(\frac{-1-i\sqrt{3}}{2}\right)b\right)^n + \left(a+\left(\frac{-1+i\sqrt{3}}{2}\right)b\right)^n$$

Where $i = \sqrt{-1}$.

Note to self: It's kinda crazy that complex numbers show up, but apparently the answer is related to a deeper fact about the trace of a matrix — that the trace of a matrix M is the sum of its eigenvalues.

Eigenvalues, in general, need not be real numbers. This is why we get complex numbers as the result.